Nuclear Particle Physics (PH509)

The Four Fundamental Forces in Nature and Their Interaction Particles

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1 Introduction

All physical phenomena in nature can be explained in terms of four fundamental interactions. These interactions govern the behavior of particles across vastly different length scales, time scales, and energy regimes. The four fundamental forces are:

1. Gravitational force
2. Electromagnetic force
3. Strong force
4. Weak force

In classical physics, forces are treated as direct interactions between objects. However, at nuclear and sub-nuclear length scales, this description becomes inadequate. Modern physics, particularly **Quantum Field Theory (QFT)**, reinterprets forces as interactions mediated by the exchange of virtual particles.

2 Concept of Force and Interaction

In classical mechanics, force is defined using Newton's second law:

$$\vec{F} = m\frac{d^2\vec{x}}{dt^2} \tag{1}$$

When potential energy is present, the total energy is:

$$E = E_k + V \tag{2}$$

At relativistic energies, particle dynamics must satisfy:

$$E = \sqrt{p^2c^2 + m_0^2c^4} \tag{3}$$

In the quantum regime, energy is quantized:

$$E_n = \left(n + \frac{1}{2}\right) h\nu \tag{4}$$

Thus, classical, relativistic, and quantum mechanics act as **languages** to describe particles subjected to fundamental forces.

In QFT, force is no longer treated as a classical field but as an interaction caused by the **exchange of mediator particles**.

3 Gravitational Interaction

Source and Nature

- Source: Mass–energy
- Affects: All particles
- Nature: Always attractive

Range and Strength

- Range: Infinite
- Relative strength: $\sim 10^{-40}$

Exchange Particle

- Graviton (hypothetical, spin-2)

Gravitation dominates at astronomical scales but is negligible at atomic and nuclear scales.

4 Electromagnetic Interaction

Source and Nature

- Source: Electric charge
- Affects: Charged particles
- Nature: Attractive and repulsive

Coupling Constant

The strength of electromagnetic interaction is characterized by the fine structure constant:

$$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c} \approx \frac{1}{137} \tag{5}$$

The effective Coulomb potential is:

$$V(r) = \frac{\alpha}{r} \tag{6}$$

Range and Time Scale

- Range: Infinite
- Characteristic time: 10^{-20} s

Exchange Particle

- Photon (γ), massless, spin-1

5 Strong Interaction

5.1 Nuclear (Residual Strong) Force

- Source: Nuclear charge
- Affects: Nucleons (p–p, p–n, n–n)
- Charge independent

Exchange Particles

- Mesons: $\pi, \rho, \omega, \delta, \phi$

Range and Strength

- Range: $\sim 10^{-15}$ m
- Characteristic time: 10^{-23} s
- Coupling constant:

$$\frac{g^2}{\hbar c} \approx 15 \tag{7}$$

5.2 Quark (Fundamental Strong) Force

- Source: Colour charge
- Exchange particles: Gluons (8)
- Framework: Quantum Chromodynamics (QCD)

6 Weak Interaction

Motivation

Weak interaction is required to explain long particle lifetimes and radioactive β-decay.

Source and Range

- Source: Weak charge
- Range: $< 10^{-17}$ m
- Characteristic time: 10^{-10} s

Exchange Particles

- W^+, W^-, Z^0 (Intermediate Vector Bosons)

Coupling Constant

$$G_w = 1.41 \times 10^{-62} \text{ J m}^3 \tag{8}$$

$$g_w \approx 5 \times 10^{-14} \tag{9}$$

7 Feynman diagram of Fundamental Interactions

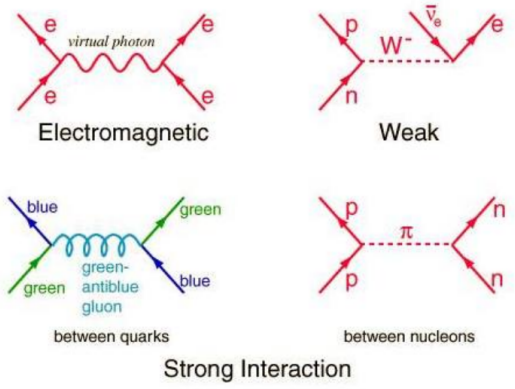


Figure 1: Feynman diagram of fundamental forces

8 Comparison of Fundamental Interactions

Interaction	Affected Particles	Range (m)	Strength	Exchange Particle
Strong	Hadrons	10^{-15}	~ 1	Mesons / Gluons
Electromagnetic	Charged	∞	10^{-2}	Photon
Weak	Hadrons & Leptons	10^{-17}	10^{-13}	$W^{\pm}, Z^0$
Gravitational	All	∞	10^{-40}	Graviton

9 Conclusion

The four fundamental forces govern nature across different scales. Their effects depend strongly on interaction range, coupling strength, and mediator particle mass. Quantum Field Theory provides a unified language in which all interactions are described as arising from the exchange of interaction particles.

Particle Classification in the Standard Model of Particle Physics

Assignment 1

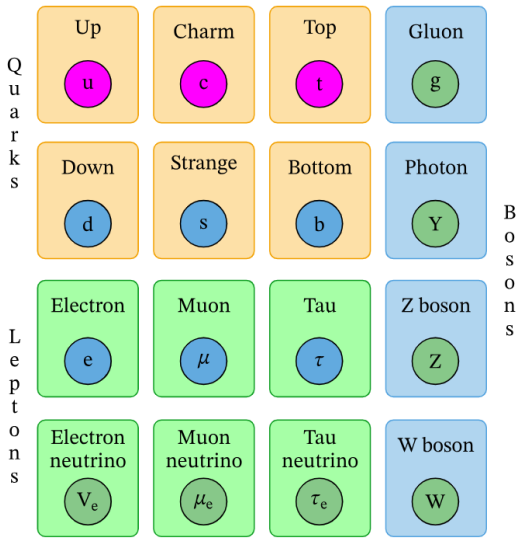
Group EE2

Introduction

In particle physics, everything in the universe is made from extremely small building blocks known as **fundamental particles**. These particles are not made of anything smaller.

The Standard Model of Particle Physics classifies all known fundamental particles into three main categories:

- **Quarks**
- **Leptons**
- **Gauge Bosons**



Each category plays a unique role in forming matter and governing the forces of nature.

Quarks

Quarks are fundamental particles that combine with one another to form composite particles such as **protons** and **neutrons**. These protons and neutrons together make up the nucleus of an atom, which means quarks are essential for the existence of matter.

Key Points about Quarks

- Quarks are never found alone in nature; they always exist in combinations.
- They interact through the **strong nuclear force**, which is one of the strongest forces in nature.
- There are six types of quarks, known as **flavors**:
  - Up (u), Down (d)
  - Charm (c), Strange (s)
  - Top (t), Bottom (b)
- Composition examples:
  - Proton consists of two up quarks and one down quark.
  - Neutron consists of one up quark and two down quarks.

Leptons

Leptons are fundamental particles that do not take part in the strong nuclear force. Unlike quarks, leptons can exist freely without combining to form other particles.

Key Points about Leptons

- There are six leptons, divided into charged leptons and neutrinos:
  - Charged leptons: **Electron, Muon, Tau**
  - Neutral leptons: **Electron Neutrino, Muon Neutrino, Tau Neutrino**
- The **electron** plays a crucial role in electricity, chemical reactions, and bonding between atoms.
- Neutrinos have extremely small mass and interact very weakly with matter, making them difficult to detect.

Gauge Bosons

Gauge bosons are fundamental particles responsible for mediating the forces between matter particles. They act as carriers that allow particles to interact with each other.

Types of Gauge Bosons

- **Photon**: Carries the electromagnetic force and is responsible for light and electrical interactions.
- **Gluon**: Carries the strong nuclear force that binds quarks together inside protons and neutrons.
- **W and Z Bosons**: Carry the weak nuclear force, which is responsible for processes such as radioactive decay.
- **Graviton**: A theoretical particle proposed to carry gravitational force; it has not yet been experimentally observed.

Assignment 1: Nuclear & Particle Physics PH509
Group ME-1

February 16, 2026

Problem Statement

Explain the basic nuclear properties and derive the expression for nuclear magnetic moment.

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1 Structure and Size of the Nucleus

An atom consists of a compact central nucleus surrounded by electrons occupying atomic orbitals. The nucleus contains all the positive charge and nearly the entire mass of the atom. It is composed of protons and neutrons, collectively referred to as nucleons. While electrons are considered structureless at present experimental limits, nucleons themselves possess internal structure and are composed of quarks. However, for nuclear physics, nucleons are treated as fundamental interacting particles.

The physical size of a nucleus is extremely small compared to that of an atom. While atomic radii are of the order of 10^{-10} m, nuclear dimensions are approximately 10^{-15} m. This length scale is known as the femtometer (fm).

Since nuclei are quantum systems, their size cannot be defined sharply. Instead, nuclear dimensions are inferred from measurements of charge or matter distributions. High-energy electron scattering experiments reveal that the nuclear radius follows the empirical relation:

$$R = R_0 A^{1/3} \tag{1}$$

where A is the mass number and $R_0 \approx 1.2$–$1.25\,\mathrm{fm}$. This relation implies that nuclear volume is proportional to the number of nucleons, indicating that nuclear matter has nearly constant density across different nuclei.

2 Binding Energy

The mass of a nucleus is not simply equal to the sum of the masses of its constituent protons and neutrons. Precision measurements using mas spectrometry show that nuclear masses are slightly smaller. This difference is called the mass defect and corresponds to the energy released when nucleons bind together.

The binding energy is:

$$B = [Zm_p + (A - Z)m_n - M(A, Z)]\, c^2 \tag{2}$$

This energy represents the amount required to completely separate a nucleus into free nucleons. A larger binding energy implies greater nuclear stability. When the binding energy per nucleon (B/A) is plotted against the mass number A, it rises rapidly for light nuclei, reaches a maximum near iron ($A \approx 56$), and then slowly decreases for heavier nuclei. This explains why energy can be released through the fusion of light nuclei and the fission of heavy nuclei.

3 Nuclear Mass

Atomic nuclei are composed of a bound system of Z protons and N neutrons, collectively called nucleons. They can be represented graphically on a plane using Z and N as coordinate axes. Most nuclei lie close to the diagonal line where $N = Z$, meaning they contain nearly equal numbers of protons and neutrons. However, only a limited number of these nuclei are stable against radioactive decay or other transformations.

For heavier elements (denoted as ${}^{A}_{Z}X$) with mass number $A \geq 100$, stability requires an excess of neutrons over protons. The most stable nuclei therefore lie along a curve slightly above the $N = Z$ line. Surrounding this region is a broad area of unstable nuclei. These unstable nuclei move toward stability through beta decay, converting:

- a neutron into a proton (β^- decay),
- a proton into a neutron (β^+ decay), or
- via electron capture.

The corresponding reactions can be written as follows:

$$ {}^{A}_{Z}X_N \rightarrow {}^{A}_{Z+1}Y_{N-1} + e^- + \bar{\nu}_e \tag{3} $$

$$ {}^{A}_{Z}X_N \rightarrow {}^{A}_{Z-1}Y_{N+1} + e^+ + \nu_e \tag{4} $$

$$ {}^{A}_{Z}X_N + e^- \rightarrow {}^{A}_{Z-1}Y_{N+1} + \nu_e \tag{5} $$

In addition to beta processes, heavier nuclei may emit alpha particles (helium nuclei) or undergo spontaneous alpha decay and nuclear fission. These processes significantly limit the region of stability among heavy nuclei.

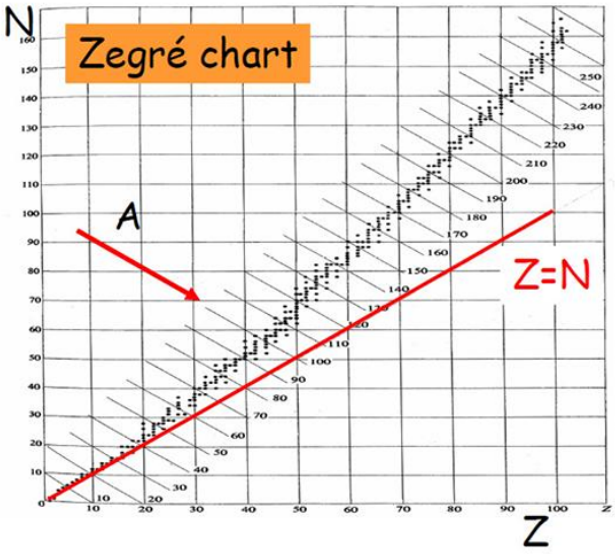


Figure 1: Zegre (N vs Z) chart showing the line of nuclear stability. Stable nuclei lie near the diagonal $N = Z$ line for light nuclei, while heavier nuclei require neutron excess.

4 Nuclear Density

The strong nuclear force exhibits saturation and charge independence, which implies that each nucleon occupies nearly the same volume inside the nucleus. As a result, nuclear matter has an almost uniform density throughout the interior of the nucleus.

Since the nuclear volume is proportional to the mass number A, the nuclear radius is given by

$$R = r_0 A^{1/3} \tag{6}$$

where $r_0 \approx 1.2$–$1.4\,\text{fm}$.This relation shows that as nucleons are added, the nucleus grows in size while its density remains nearly constant.

The spatial distribution of nuclear matter or charge is well described by the **Fermi (Woods–Saxon) distribution**:

$$\rho(r) = \frac{\rho_0}{1 + e^{(r-R_0)/a}} \tag{7}$$

where ρ_0 is the central density, R_0 is the half-density radius, and α represents surface diffuseness. This indicates a flat density in the interior and a gradual fall near the surface.

Experimentally, nuclear density is determined using elastic scattering of electrons and nucleons. These measurements confirm that central nuclear density is almost the same for all nuclei. The average nuclear matter density is approximately

$$\rho \approx 1.4 \times 10^{17}\,\text{kg m}^{-3} \tag{8}$$

which is independent of A and about 10^{14} times greater than ordinary matter density, demonstrating the highly compact nature of nuclei.

5 Nuclear Angular Momentum and Parity

Each nucleon inside a nucleus has orbital angular momentum l and intrinsic spin s, which combine to give its total angular momentum j. The total angular momentum of the nucleus, called the nuclear spin and denoted by I, is obtained by adding the angular momenta of all nucleons. In most nuclei, nucleons pair up and cancel each other's angular momentum.

Therefore, the nuclear spin is often determined by the unpaired nucleon, giving $I = j$. Nuclei with an even mass number have integer spin, while those with an odd mass number have half-integer spin. Most even–even nuclei have ground state spin $I = 0$, showing strong nucleon pairing. Parity (π)describes the symmetry of the nuclear wave function and can be positive ($+$) or negative ($-$). It is treated as an overall property of the nucleus and is determined experimentally.

Nuclear states are labeled by spin and parity, written as I^{π}, such as $0^{+}, 1/2^{+}$ or $5/2^{+}$. There is no fixed relation between spin and parity; any spin value can have either parity.

6 Nuclear Moments

Nuclear moments arise due to the motion and intrinsic properties of protons and neutrons inside the nucleus. Since protons carry charge and both nucleons possess angular momentum and spin, nuclei exhibit electromagnetic multipole moments. The two most important are the magnetic dipole moment and the electric quadrapole moment.

6.1 Magnetic Dipole Moment

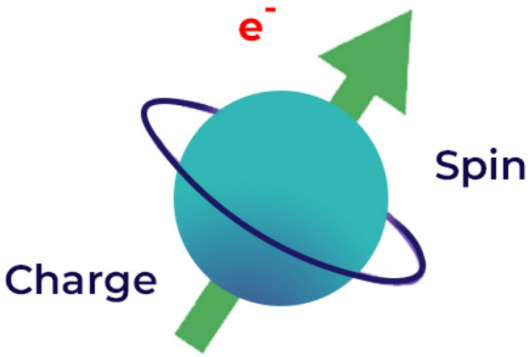


Figure 2: Electron Magnetic Moment due to Spin

An electron has intrinsic spin S and electric charge e. Therefore, it possesses a magnetic moment μ_s.

For angular momentum,

$$L = m.v.r$$

Magnetic moment due to a current loop:

$$\mu_L = \text{current} \times \text{area}$$

$$I = \frac{e}{T}, \qquad A = \pi r^2$$

Since

$$v = \frac{2\pi r}{T}$$

$$\Rightarrow \frac{\mu_L}{L} = \frac{\frac{e}{T}\pi r^2}{mvr} = \frac{e}{2m}$$

Thus,

$$\mu_L = \frac{e}{2m}L$$

In microscopic quantum scale, since $L = \#\hbar$,

$$\mu_L = \left(\frac{e\hbar}{2m}\right)\left(\frac{L}{\hbar}\right)$$

Let's say

$$\mu_B = \frac{e\hbar}{2m}$$

This μ_B is known as the **Bohr magneton**.

$$\therefore \quad \mu_L = \mu_B\left(\frac{L}{\hbar}\right)$$

Similarly for spin,

$$\frac{\mu_s}{S} = g_s\left(\frac{e}{2m}\right)$$

$$\Rightarrow \mu_s = g_s\mu_B\left(\frac{S}{\hbar}\right)$$

where $g_s = 2$ is the spin g-factor and $S = \#\hbar$ is spin angular momentum of electron.

Similar to the electron, the proton has a spin magnetic moment:

$$\mu_s = 2\beta_n\left(\frac{S}{\hbar}\right)$$

where

$$\beta_n = \frac{e\hbar}{2m_p}$$

This β_n is called the **nuclear magneton** (same as Bohr magneton quantity but for proton).

$$\left.\begin{array}{l}\text{Spin angular moment of proton is } \mu_p = +2.792\,\beta_n \\ \text{Spin angular moment of neutron is } \mu_n = -1.913\,\beta_n\end{array}\right\} \quad \boxed{\text{Why?}}$$

Because protons and neutrons are not elementary particles, they are made of quarks.

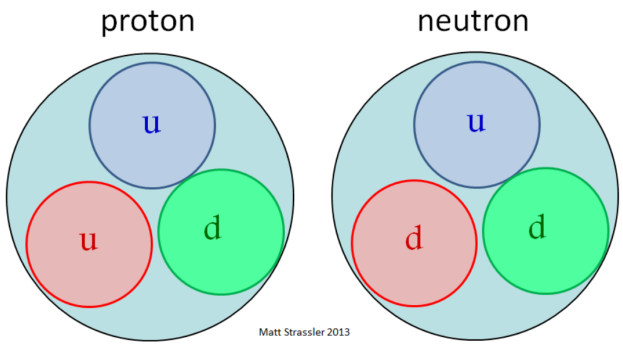


Figure 3: Internal Quark Structure

Their internal quark structure causes their magnetic moments to differ from the simple Dirac prediction.

6.2 Electric Quadrupole Moment

If the nuclear charge distribution is not spherical, it produces a quadrupole moment. The quadrupole tensor is:

$$Q_{ij} = \int \rho(\mathbf{r}) \left(3x_i x_j - r^2 \delta_{ij}\right) d\tau \tag{9}$$

For axial symmetry:

$$Q = \int \rho(\mathbf{r})(3z^2 - r^2) d\tau \tag{10}$$

Interpretation:

- $Q = 0 \rightarrow$ Spherical nucleus
- $Q > 0 \rightarrow$ Prolate (elongated)
- $Q < 0 \rightarrow$ Oblate (flattened)

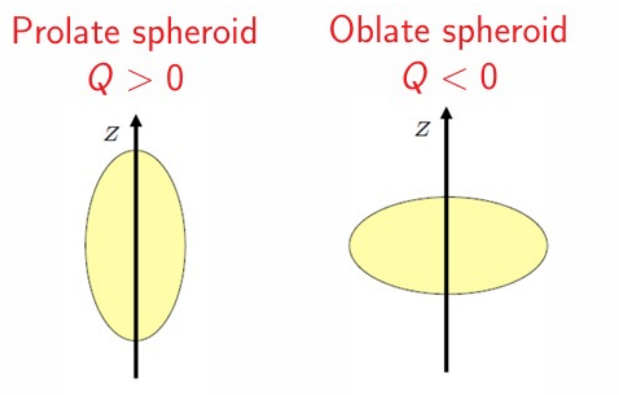


Figure 4: Prolate and oblate nuclear shapes ($Q > 0$, $Q < 0$).

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ASSIGNMENT

Course Details:

Course : Nuclear and Particle Physics
Course Code : PHL509/PH509
Instructor : Dr. Sabyasachi Ghosh
Batch : CS1

Question :

The nucleus can be modelled as a degenerate gas of fermions. Write the thermal distribution and obtain the grand potential and the number density.

Fermi Gas Model of the Nucleus

Introduction

In this model, the nucleus is treated as a **degenerate gas of non-interacting fermions (nucleons)** confined inside a sphere of radius

$$R = r_0 A^{1/3}, \qquad r_0 \approx 1.2 \text{ fm}$$

The volume of the nucleus is

$$V = \frac{4}{3}\pi r_0^3 A$$

Each quantum state can hold two nucleons (spin up and spin down). Because nuclear temperature is extremely small compared to Fermi temperature, the system is strongly degenerate.

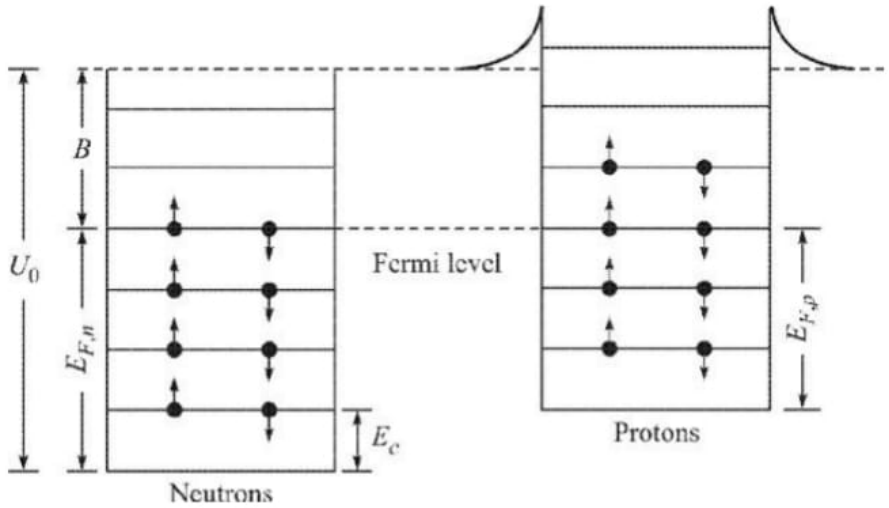


Figure Protons–neutrons levels in Fermi model of the nucleus. The two potential wells have slightly different shapes because of the Coulomb repulsion of protons.

1. Thermal Distribution (Fermi–Dirac Distribution)

Step 1: Grand Canonical Ensemble

In the grand canonical ensemble, the probability of a system being in a state with energy E and particle number N is proportional to

$$e^{-\beta(E-\mu N)}$$

where

$$\beta = \frac{1}{k_B T}$$

and μ is the chemical potential.

Step 2: Single Particle State

For a single quantum state of energy ϵ:

Since nucleons are fermions, occupation number can only be

$$n = 0 \quad \text{or} \quad n = 1$$

The grand partition function for this single state is:

$$Z = \sum_{n=0}^{1} e^{-\beta(\epsilon n - \mu n)}$$

Substitute values:

For $n = 0$:

$$e^0 = 1$$

For $n = 1$:

$$e^{-\beta(\epsilon - \mu)}$$

Thus,

$$Z = 1 + e^{-\beta(\epsilon - \mu)}$$

Step 3: Average Occupation Number

The average occupation number is

$$f(\epsilon) = \langle n \rangle = \frac{\sum n e^{-\beta(\epsilon n - \mu n)}}{Z}$$

Substituting:

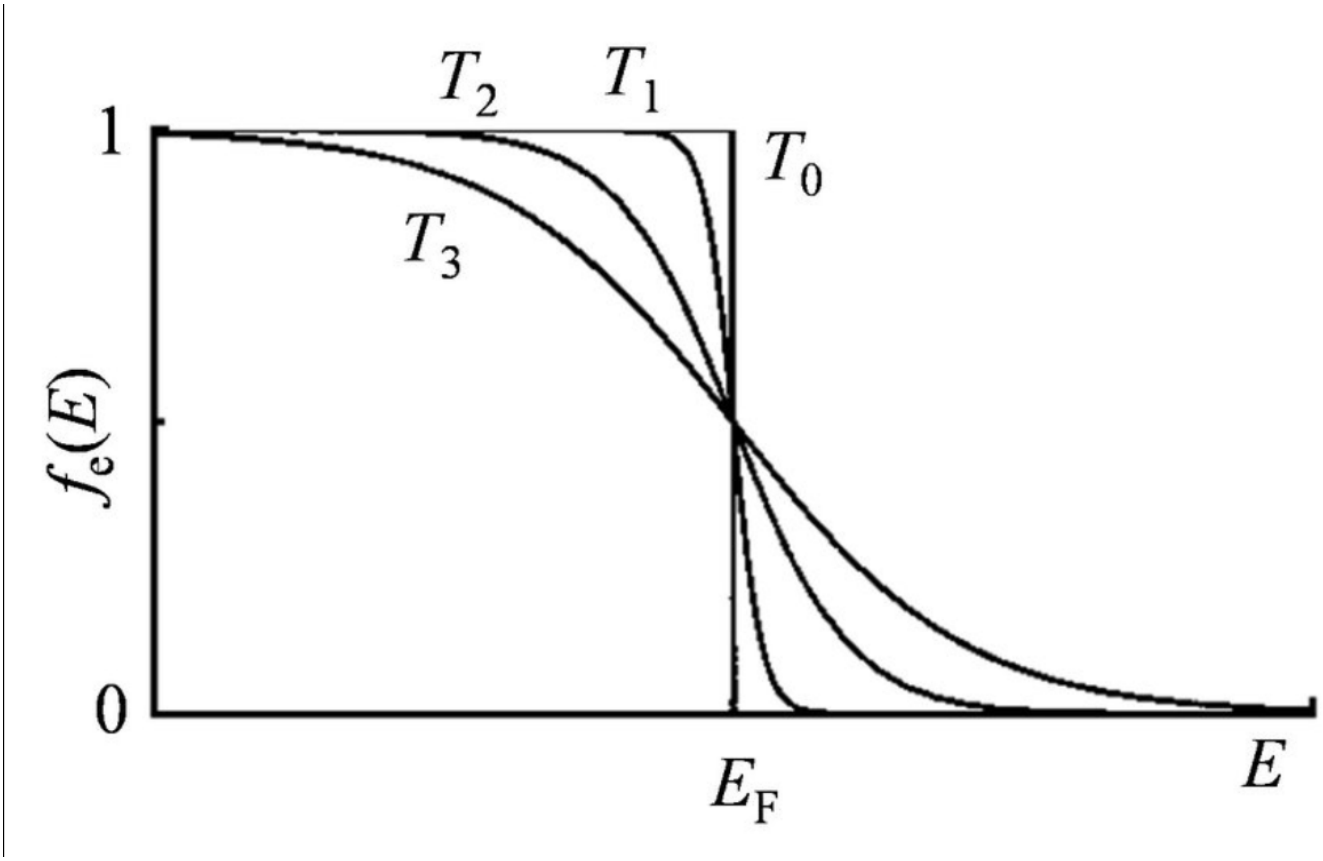
$$f(\epsilon) = \frac{0 \cdot 1 + 1 \cdot e^{-\beta(\epsilon - \mu)}}{1 + e^{-\beta(\epsilon - \mu)}}$$

$$f(\epsilon) = \frac{e^{-\beta(\epsilon-\mu)}}{1 + e^{-\beta(\epsilon-\mu)}}$$

Multiply numerator and denominator by $e^{\beta(\epsilon-\mu)}$:

$$\boxed{f(\epsilon) = \frac{1}{e^{\beta(\epsilon-\mu)} + 1}}$$

This is the Fermi–Dirac thermal distribution.



2. Grand Potential

Step 1: Grand Partition Function

For many independent states:

$$Z_G = \prod_p \left(1 + e^{-\beta(\epsilon_p-\mu)}\right)$$

Take logarithm:

$$\ln Z_G = \sum_p \ln\left(1 + e^{-\beta(\epsilon_p - \mu)}\right)$$

Step 2: Definition of Grand Potential

Grand potential is

$$\Omega = -k_B T \ln Z_G$$

Substitute:

$$\boxed{\Omega = -k_B T \sum_p \ln\left(1 + e^{-\beta(\epsilon_p - \mu)}\right)}$$

Step 3: Convert Summation to Integral

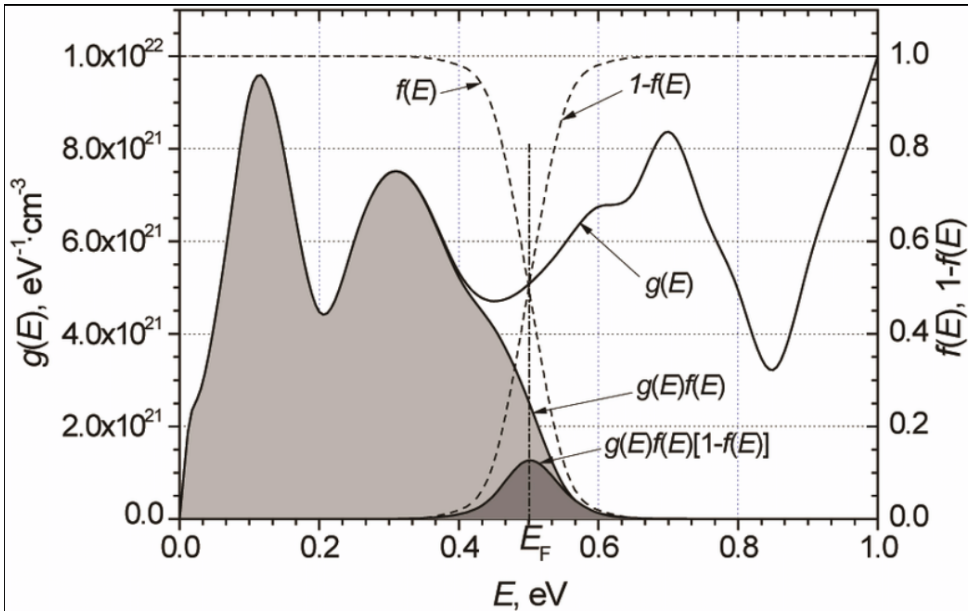
In phase space, number of states in momentum space is:

$$\sum_p \to \frac{2V}{h^3} \int d^3p$$

(2 is spin degeneracy)

Thus,

$$\boxed{\Omega = -\frac{2V k_B T}{h^3} \int d^3p \ln\left(1 + e^{-\beta(\epsilon - \mu)}\right)}$$



3. Number Density

Step 1: Thermodynamic Relation

$$N = -\left(\frac{\partial \Omega}{\partial \mu}\right)_{T,V}$$

Substitute expression of Ω:

$$N = -\frac{\partial}{\partial \mu}\left[-\frac{2Vk_BT}{h^3}\int d^3p \ln\left(1 + e^{-\beta(\epsilon-\mu)}\right)\right]$$

$$N = \frac{2Vk_BT}{h^3}\int d^3p \frac{\partial}{\partial \mu}\ln\left(1 + e^{-\beta(\epsilon-\mu)}\right)$$

Step 2: Differentiate Logarithm

Let

$$x = 1 + e^{-\beta(\epsilon-\mu)}$$

$$\frac{d}{d\mu}\ln x = \frac{1}{x}\frac{dx}{d\mu}$$

Now,

$$\frac{dx}{d\mu} = e^{-\beta(\epsilon-\mu)}\beta$$

Therefore,

$$\frac{\partial}{\partial \mu}\ln\left(1 + e^{-\beta(\epsilon-\mu)}\right) = \frac{\beta e^{-\beta(\epsilon-\mu)}}{1 + e^{-\beta(\epsilon-\mu)}}$$

Substitute into N:

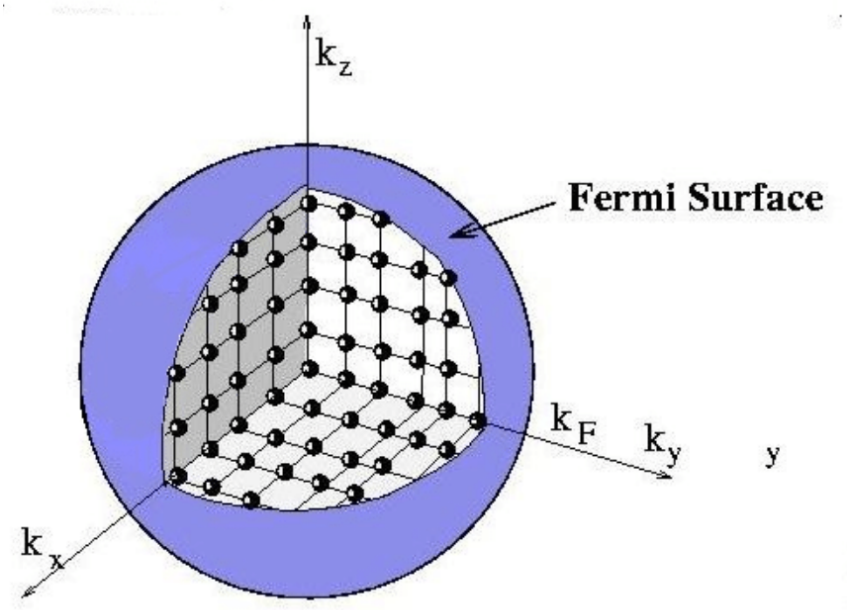
$$N = \frac{2Vk_BT}{h^3}\int d^3p \frac{\beta e^{-\beta(\epsilon-\mu)}}{1 + e^{-\beta(\epsilon-\mu)}}$$

Since $\beta k_B T = 1$:

$$\boxed{N = \frac{2V}{h^3}\int d^3p \frac{1}{e^{\beta(\epsilon-\mu)} + 1}}$$

Divide by V:

$$\boxed{n = \frac{N}{V} = \frac{2}{h^3}\int d^3p \frac{1}{e^{\beta(\epsilon-\mu)}+1}}$$



4. Degenerate Limit ($T = 0$)

At $T = 0$:

$$f(\epsilon) = \begin{cases} 1 & \epsilon \leq E_F \\ 0 & \epsilon > E_F \end{cases}$$

Step 1: Number Density

$$n = \frac{2}{h^3}\int_{p \leq P_F} d^3p$$

Convert to spherical coordinates:

$$d^3p = 4\pi p^2 dp$$

$$n = \frac{2}{h^3} 4\pi \int_0^{P_F} p^2 dp$$

$$= \frac{8\pi}{h^3}\left[\frac{P_F^3}{3}\right]$$

$$n = \frac{8\pi}{3h^3} P_F^3$$

Using $h = 2\pi\hbar$:

$$h^3 = 8\pi^3\hbar^3$$

$$\boxed{n = \frac{P_F^3}{3\pi^2\hbar^3}}$$

Step 2: Fermi Momentum

Rearranging:

$$\boxed{P_F = \hbar(3\pi^2 n)^{1/3}}$$

Step 3: Nuclear Application

Using

$$V = \frac{4}{3}\pi r_0^3 A$$

For symmetric nucleus $N = Z = A/2$:

$$P_F \approx 260 - 270\,\mathrm{MeV}/c$$

Fermi energy:

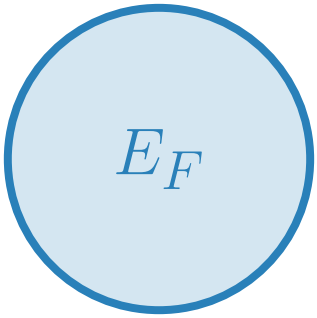
$$E_F = \frac{P_F^2}{2m} \approx 33\,\mathrm{MeV}$$

This confirms the nucleus behaves as a strongly degenerate Fermi gas.

Nuclear Physics

Assignment 1

Degenerate Fermi Gas



Question Statement

The nucleus can be modelled as a degenerate gas of fermions. Obtain the expressions for **number density**, **energy density** and **average energy** considering a non-relativistic energy dispersion relation:

$$E = \frac{p^2}{2m}$$

Group CS2

February 2026

What is a Degenerate Fermi Gas?

Definition

A **degenerate Fermi gas** is a system of fermions (particles with half-integer spin) that obey the **Pauli Exclusion Principle**. At zero temperature ($T = 0$), all available quantum states are filled up to a maximum energy called the **Fermi energy** E_F.

Key Properties

- Each quantum state can be occupied by at most **one fermion** (per spin state)
- All states with energy $E < E_F$ are **completely filled**
- All states with energy $E > E_F$ are **completely empty**

Physical Interpretation

This situation is called **degeneracy** because the behavior of the system is dominated by *quantum statistics* rather than temperature. The quantum nature of the particles determines their distribution, not thermal effects.

Nuclear Physics Application

Why Can the Nucleus be Treated as a Degenerate Fermi Gas?

Important Concept

The nucleus consists of **nucleons** (protons and neutrons), which are:

- Fermions (spin $\frac{1}{2}$ particles)
- Confined in a very small volume ($\sim 10^{-15}$ m)
- Present at extremely high density

Conditions Inside the Nucleus

Inside the nucleus:

- Thermal energies ($k_BT \sim 1$ MeV at nuclear temperatures) are much smaller than the Fermi energy ($E_F \sim$ 30-40 MeV)
- Nucleons behave approximately as a **zero-temperature quantum system**
- Quantum degeneracy pressure provides stability against gravitational collapse

Physical Interpretation

Therefore, the nucleus can be approximated as a **degenerate gas of non-relativistic fermions**, similar to the electron gas in white dwarf stars or the conduction electrons in metals at low temperatures.

Mathematical Framework

Non-Relativistic Dispersion Relation

We assume that nucleons are **non-relativistic** particles, so:

$$\boxed{E = \frac{p^2}{2m}}$$

where:

- E = kinetic energy of the particle
- p = magnitude of momentum
- m = mass of nucleon (~ 938 MeV/c^2)

Important Variables

Symbol	Quantity
g	Spin degeneracy factor (for spin-$\frac{1}{2}$: $g = 2$)
h	Planck's constant (6.626×10^{-34} J·s)
p_F	Fermi momentum (maximum momentum at $T = 0$)
E_F	Fermi energy (maximum energy at $T = 0$)
n	Number density (particles per unit volume)
ε	Energy density (energy per unit volume)
$\bar{E}$	Average energy per particle

Derivation of Number Density

Starting Point: Density of States

The number of quantum states in momentum space is given by:

$$dn = g\frac{d^3p}{h^3}$$

where the factor h^3 comes from the quantum mechanical volume of a single state in phase space.

In **spherical coordinates**, the volume element in momentum space is:

$$d^3p = 4\pi p^2\, dp$$

This accounts for all directions in 3D momentum space at radius p.

At $T = 0$, all states are filled up to the Fermi momentum p_F, so we integrate:

$$n = g\int_0^{p_F} \frac{4\pi p^2\, dp}{h^3}$$

Step-by-Step Calculation

Detailed Calculation

Step 1: Extract constants from the integral

$$n = g\int_0^{p_F} \frac{4\pi p^2\, dp}{h^3}$$

$$= \frac{g\cdot 4\pi}{h^3}\int_0^{p_F} p^2\, dp$$

Step 2: Evaluate the integral using $\int p^2\,dp = \frac{p^3}{3}$

$$n = \frac{g \cdot 4\pi}{h^3}\left[\frac{p^3}{3}\right]_0^{p_F}$$

$$= \frac{g \cdot 4\pi}{h^3} \cdot \frac{p_F^3}{3}$$

$$= \frac{4\pi g\, p_F^3}{3h^3}$$

Final Result

General Result:

$$\boxed{n = \frac{4\pi g\, p_F^3}{3h^3}}$$

For spin-$\frac{1}{2}$ particles ($g = 2$):

$$\boxed{n = \frac{8\pi p_F^3}{3h^3}}$$

Derivation of Energy Density

Definition and Setup

The **energy density** is the total energy per unit volume. Each state contributes energy $E(p) = \frac{p^2}{2m}$:

$$\varepsilon = g \int_0^{p_F} \frac{4\pi p^2 \, dp}{h^3} \cdot \frac{p^2}{2m}$$

This can be rewritten as:

$$\varepsilon = \frac{g \cdot 4\pi}{h^3 \cdot 2m} \int_0^{p_F} p^4 \, dp$$

Complete Derivation

Detailed Calculation

Step 1: Combine the energy factor with density of states

$$\varepsilon = g \int_0^{p_F} \frac{4\pi p^2 \, dp}{h^3} \cdot \frac{p^2}{2m}$$

$$= \frac{g \cdot 4\pi}{h^3 \cdot 2m} \int_0^{p_F} p^4 \, dp$$

Step 2: Simplify constants

$$\varepsilon = \frac{g \cdot 4\pi}{2h^3 m} \int_0^{p_F} p^4 \, dp$$

$$= \frac{2\pi g}{h^3 m} \int_0^{p_F} p^4 \, dp$$

Step 3: Evaluate the integral using $\int p^4\,dp = \frac{p^5}{5}$

$$\varepsilon = \frac{2\pi g}{h^3 m}\left[\frac{p^5}{5}\right]_0^{p_F}$$

$$= \frac{2\pi g}{h^3 m} \cdot \frac{p_F^5}{5}$$

$$= \frac{2\pi g\, p_F^5}{5h^3 m}$$

Final Result

General Result:

$$\boxed{\varepsilon = \frac{2\pi g\, p_F^5}{5h^3 m}}$$

For spin-$\frac{1}{2}$ particles ($g = 2$):

$$\boxed{\varepsilon = \frac{4\pi p_F^5}{5h^3 m}}$$

Average Energy Per Particle

Definition

The average energy per particle is simply the ratio of total energy to total number of particles:

$$\bar{E} = \frac{\text{Total Energy}}{\text{Total Number of Particles}} = \frac{\varepsilon}{n}$$

Calculation

Detailed Calculation

Step 1: Substitute our expressions for ε and n

$$\bar{E} = \frac{\varepsilon}{n}$$

$$= \frac{\frac{2\pi g\, p_F^5}{5h^3 m}}{\frac{4\pi g\, p_F^3}{3h^3}}$$

Step 2: Simplify by multiplying by the reciprocal

$$\bar{E} = \frac{2\pi g\, p_F^5}{5h^3 m} \times \frac{3h^3}{4\pi g\, p_F^3}$$

$$= \frac{6\pi g h^3\, p_F^5}{20\pi g h^3 m\, p_F^3}$$

$$= \frac{6\, p_F^2}{20\, m}$$

Step 3: Simplify the fraction

$$\bar{E} = \frac{6\, p_F^2}{20\, m}$$

$$= \frac{3\, p_F^2}{10\, m}$$

Step 4: Relate to the Fermi energy
Recall that the Fermi energy is defined as:

$$E_F = \frac{p_F^2}{2m}$$

Therefore:

$$p_F^2 = 2mE_F$$

Substituting back:

$$\begin{aligned}\bar{E} &= \frac{3\,p_F^2}{10\,m} \\ &= \frac{3}{10m} \cdot 2mE_F \\ &= \frac{6m\,E_F}{10m} \\ &= \frac{3E_F}{5}\end{aligned}$$

Final Result

Final Result:

$$\boxed{\bar{E} = \frac{3}{5}E_F}$$

Physical Interpretation

This remarkable result shows that the **average energy** of particles in a degenerate Fermi gas is $\frac{3}{5}$ of the **maximum energy** (Fermi energy). This is because the energy distribution is not uniform—many particles occupy lower energy states, bringing down the average.

Summary of Results

Final Formulas

Quantity	Formula
Number Density	$n = \frac{4\pi g\, p_F^3}{3h^3}$
Energy Density	$\varepsilon = \frac{2\pi g\, p_F^5}{5h^3 m}$
Average Energy	$\bar{E} = \frac{3}{5} E_F$
Fermi Energy	$E_F = \frac{p_F^2}{2m}$

For spin-$\frac{1}{2}$ particles (nucleons), use $g = 2$

Physical Interpretation

Physical Significance

These formulas are fundamental to understanding:

- **Nuclear Structure:** The density and energy of nucleons in atomic nuclei
- **White Dwarfs:** Electron degeneracy pressure supporting stars
- **Neutron Stars:** Neutron degeneracy in extreme conditions
- **Condensed Matter:** Free electron gas in metals

The degenerate Fermi gas model is one of the most important concepts in quantum statistical mechanics, bridging nuclear physics, astrophysics, and solid-state physics.

END OF DOCUMENT

Indian Institute of Technology Bhilai

ASSIGNMENT

Course	:	Nuclear and Particle Physics
Course Code	:	PHL509/PH509
Instructor	:	Dr. Sabyasachi Ghosh
Batch	:	DSAI+MSME

Question:

The nucleus can be modelled as a degenerate gas of fermions. Obtain the expressions for number density, energy density and average energy considering a relativistic energy dispersion relation $E = \sqrt{p^2c^2 + m^2c^4}$.

1 Introduction to Fermi Gas Model

The Fermi gas model is a fundamental theoretical framework for understanding the nuclear structure. In this model, the nucleus is treated as a **degenerate gas of non-interacting fermions (nucleons)** confined within a spherical potential well of radius:

$$R = r_0 A^{1/3}, \quad r_0 \approx 1.2 \text{ fm} \tag{1}$$

where A is the mass number. The nuclear volume is given by:

$$V = \frac{4}{3}\pi r_0^3 A \tag{2}$$

1.1 Key Features

- **Degeneracy:** Nuclear temperatures ($T \sim 1$ MeV) are extremely small compared to the Fermi temperature ($T_F \sim 10^{12}$ K), making the system strongly degenerate.
- **Pauli Exclusion:** Each quantum state can hold at most two nucleons (spin up and spin down).
- **Separate Wells:** Protons and neutrons occupy separate potential wells due to the Coulomb repulsion affecting only protons.

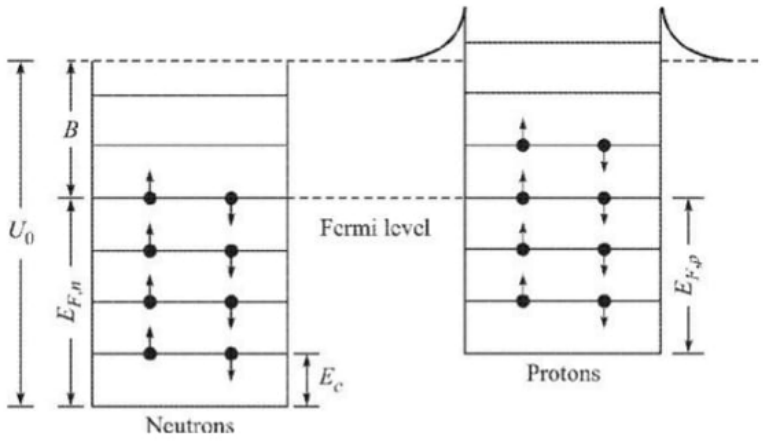


Figure 1: Energy level diagram showing the Fermi surface for protons and neutrons in the Fermi gas model. The Fermi level E_F separates occupied and unoccupied states at $T = 0$.

1.2 Relativistic Consideration

In the nucleus, nucleons have momenta $p_F \sim 260$ MeV/c, which is comparable to the nucleon rest mass $m_N c \sim 940$ MeV. Therefore, we must use the **relativistic energy-momentum relation**:

$$E = \sqrt{p^2c^2 + m^2c^4} \tag{3}$$

This contrasts with the non-relativistic approximation $E = \frac{p^2}{2m}$ used in the simpler treatment.

1.3 Thermal Distribution at $T = 0$

At zero temperature (degenerate limit), the Fermi-Dirac distribution becomes a step function:

$$f(\epsilon) = \begin{cases} 1 & \epsilon \leq E_F \\ 0 & \epsilon > E_F \end{cases} \tag{4}$$

All states up to the Fermi energy E_F are completely filled, and all states above are empty.

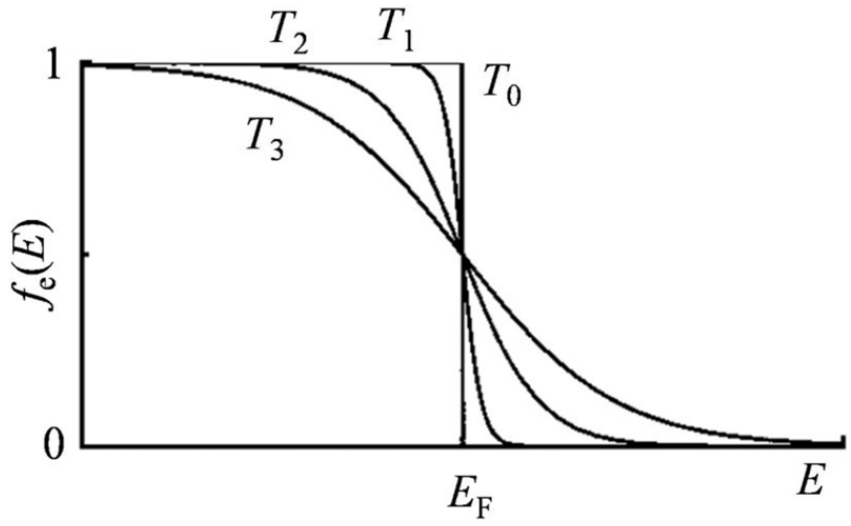


Figure 2: Fermi-Dirac distribution at various temperatures. At $T = 0$, it becomes a perfect step function at E_F.

2 Number Density with Relativistic Dispersion

2.1 Derivation from Phase Space Integration

The number density n represents the number of nucleons per unit volume. At zero temperature, all states up to the Fermi momentum p_F are occupied, and all states above are empty.

The total number of particles is obtained by integrating over phase space:

$$N = \int \frac{d^3p}{(2\pi\hbar)^3} \times g_s \times \Theta(p_F - p) \tag{5}$$

where:

- $g_s = 2$ is the spin degeneracy factor (accounts for spin up and spin down states)
- $\Theta(p_F - p)$ is the step function: $\Theta(x) = 1$ for $x > 0$, and 0 otherwise
- $(2\pi\hbar)^3$ is the volume of a single quantum state in phase space (from Heisenberg uncertainty)

Converting to spherical coordinates in momentum space, we have $d^3p = 4\pi p^2 dp$ (the angular integral gives 4π):

$$N = \frac{2}{(2\pi\hbar)^3} \int_0^{p_F} 4\pi p^2 \, dp \tag{6}$$

Evaluating the integral:

$$N = \frac{8\pi}{(2\pi\hbar)^3} \int_0^{p_F} p^2 \, dp = \frac{8\pi}{(2\pi\hbar)^3} \cdot \left[\frac{p^3}{3}\right]_0^{p_F} = \frac{8\pi}{(2\pi\hbar)^3} \cdot \frac{p_F^3}{3} \tag{7}$$

Simplifying:

$$N = \frac{8\pi p_F^3}{3 \cdot 8\pi^3\hbar^3} = \frac{p_F^3}{3\pi^2\hbar^3} \tag{8}$$

The number density is:

$$\boxed{n = \frac{N}{V} = \frac{p_F^3}{3\pi^2\hbar^3}} \tag{9}$$

Inverting this relation gives the Fermi momentum:

$$p_F = \hbar(3\pi^2 n)^{1/3} \tag{10}$$

2.2 Physical Interpretation

This result is **independent of the energy dispersion relation** because it only depends on counting the number of available momentum states up to p_F. The phase space volume element d^3p does not involve the energy-momentum relation. Whether the system is relativistic or non-relativistic does not affect the number density formula.

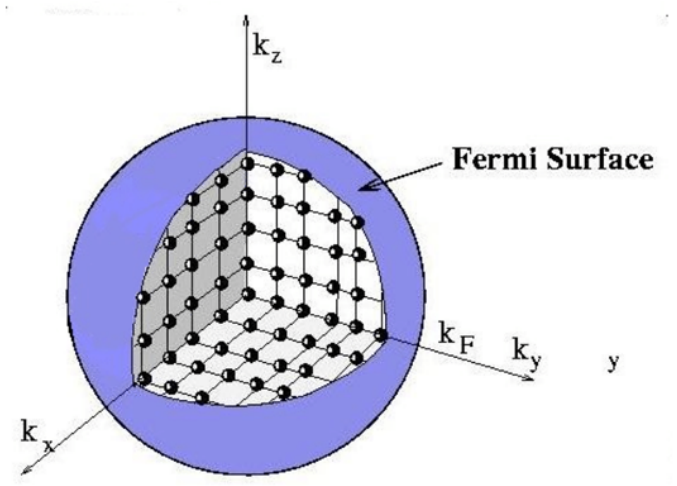


Figure 3: The Fermi sphere in momentum space. All states with $|\mathbf{p}| < p_F$ are occupied (shaded region), representing all the nucleons in the ground state.

2.3 Numerical Values for Nuclear Matter

For symmetric nuclear matter with $N = Z = A/2$ and using nuclear density $n_0 \approx 0.17$ fm^{-3}:

$$p_F = \hbar(3\pi^2 n_0)^{1/3} \tag{11}$$
$$= 197.3 \text{ MeV} \cdot \text{fm} \times (3\pi^2 \times 0.17 \text{ fm}^{-3})^{1/3} \tag{12}$$
$$\approx 260 \text{ MeV/c} \tag{13}$$

3 Energy Density with Relativistic Dispersion

3.1 Definition and Setup

The energy density ε is the total energy per unit volume. Using the relativistic dispersion relation $E(p) = \sqrt{p^2c^2 + m^2c^4}$, we must integrate the energy of all occupied states:

$$\mathcal{E} = \int \frac{d^3p}{(2\pi\hbar)^3} \times g_s \times E(p) \times \Theta(p_F - p) \tag{14}$$

3.2 Derivation

Converting to spherical coordinates:

$$\mathcal{E} = \frac{2}{(2\pi\hbar)^3} \int_0^{p_F} 4\pi p^2 \sqrt{p^2c^2 + m^2c^4}\, dp \tag{15}$$

Simplifying:

$$\mathcal{E} = \frac{8\pi}{(2\pi)^3\hbar^3} \int_0^{p_F} p^2 \sqrt{p^2c^2 + m^2c^4}\, dp = \frac{1}{\pi^2\hbar^3} \int_0^{p_F} p^2 \sqrt{p^2c^2 + m^2c^4}\, dp \tag{16}$$

3.3 Detailed Evaluation of the Integral

We need to evaluate:

$$I = \int_0^{p_F} p^2 \sqrt{p^2c^2 + m^2c^4}\, dp \tag{17}$$

Step 1: Integration by Parts
Let $u = p^2$ and $dv = p\sqrt{p^2c^2 + m^2c^4}\, dp$
Then:

$$du = 2p\, dp \tag{18}$$
$$v = \int p\sqrt{p^2c^2 + m^2c^4}\, dp \tag{19}$$

To find v, use substitution $w = p^2c^2 + m^2c^4$, so $dw = 2pc^2\, dp$:

$$v = \int p\sqrt{w}\, \frac{dw}{2pc^2} = \frac{1}{2c^2} \int \sqrt{w}\, dw = \frac{1}{2c^2} \cdot \frac{2}{3} w^{3/2} = \frac{1}{3c^2}(p^2c^2 + m^2c^4)^{3/2} \tag{20}$$

Using integration by parts:

$$I = \left[p^2 \cdot \frac{1}{3c^2}(p^2c^2 + m^2c^4)^{3/2} \right]_0^{p_F} - \int_0^{p_F} 2p \cdot \frac{1}{3c^2}(p^2c^2 + m^2c^4)^{3/2}\, dp \tag{21}$$

The first term evaluates to:

$$\left[\frac{p^2}{3c^2}(p^2c^2 + m^2c^4)^{3/2} \right]_0^{p_F} = \frac{p_F^2}{3c^2}(p_F^2c^2 + m^2c^4)^{3/2} = \frac{p_F^2}{3c^2}E_F^3 \tag{22}$$

where $E_F = \sqrt{p_F^2c^2 + m^2c^4}$.

Step 2: Evaluate the Second Integral

For the second integral:

$$I_2 = \int_0^{p_F} \frac{2p}{3c^2}(p^2c^2 + m^2c^4)^{3/2}\, dp \tag{23}$$

Using substitution $w = p^2c^2 + m^2c^4$, so $dw = 2pc^2\, dp$:

$$I_2 = \frac{2}{3c^2} \int_{m^2c^4}^{p_F^2c^2+m^2c^4} w^{3/2} \frac{dw}{2c^2} = \frac{1}{3c^4} \left[\frac{2}{5} w^{5/2} \right]_{m^2c^4}^{p_F^2c^2+m^2c^4} \tag{24}$$

$$I_2 = \frac{2}{15c^4} \left[(p_F^2c^2 + m^2c^4)^{5/2} - (m^2c^4)^{5/2} \right] = \frac{2}{15c^4} \left[E_F^5 - (mc^2)^5 \right] \tag{25}$$

Step 3: Combine the Terms

$$I = \frac{p_F^2 E_F^3}{3c^2} - \frac{2}{15c^4} \left[E_F^5 - (mc^2)^5 \right] \tag{26}$$

To combine these, factor out common terms. Multiply the first term by $\frac{5E_F^2}{5E_F^2}$:

$$I = \frac{5p_F^2 E_F^5}{15c^2 E_F^2} - \frac{2E_F^5}{15c^4} + \frac{2(mc^2)^5}{15c^4} \tag{27}$$

Note that $p_F^2 = \frac{E_F^2 - m^2c^4}{c^2}$, so:

$$\frac{p_F^2}{c^2} = \frac{E_F^2 - m^2c^4}{c^4} \tag{28}$$

After algebraic manipulation:

$$I = \frac{1}{15c^4} \left[5p_F^2c^2E_F^3 - 2E_F^5 + 2(mc^2)^5 \right] \tag{29}$$

Using $p_F^2c^2 = E_F^2 - m^2c^4$:

$$I = \frac{1}{15c^4} \left[5(E_F^2 - m^2c^4)E_F^3 - 2E_F^5 + 2(mc^2)^5 \right] \tag{30}$$

$$I = \frac{1}{15c^4} \left[5E_F^5 - 5m^2c^4E_F^3 - 2E_F^5 + 2(mc^2)^5 \right] \tag{31}$$

$$I = \frac{1}{15c^4} \left[3E_F^5 - 5m^2c^4E_F^3 + 2(mc^2)^5 \right] \tag{32}$$

This can be rewritten in a more standard form using hyperbolic functions. After further manipulation, this leads to:

$$I = \frac{1}{8}\left[p_F E_F(2p_F^2c^2 + m^2c^4) - (mc^2)^4 \sinh^{-1}\left(\frac{p_F c}{mc^2}\right)\right] \tag{33}$$

Therefore, the energy density is:

$$\boxed{\varepsilon = \frac{1}{8\pi^2\hbar^3}\left[p_F E_F(2p_F^2c^2 + m^2c^4) - (mc^2)^4 \sinh^{-1}\left(\frac{p_F c}{mc^2}\right)\right]} \tag{34}$$

3.4 Alternative Form Using Logarithm

The inverse hyperbolic sine function can be written as:

$$\sinh^{-1}(x) = \ln\left(x + \sqrt{x^2+1}\right) \tag{35}$$

Applying this with $x = \frac{p_F c}{mc^2}$:

$$\sinh^{-1}\left(\frac{p_F c}{mc^2}\right) = \ln\left(\frac{p_F c}{mc^2} + \sqrt{\frac{p_F^2c^2}{m^2c^4} + 1}\right) = \ln\left(\frac{p_F c + \sqrt{p_F^2c^2 + m^2c^4}}{mc^2}\right) \tag{36}$$

Since $E_F = \sqrt{p_F^2c^2 + m^2c^4}$:

$$\sinh^{-1}\left(\frac{p_F c}{mc^2}\right) = \ln\left(\frac{p_F c + E_F}{mc^2}\right) \tag{37}$$

Therefore, the alternative form is:

$$\boxed{\varepsilon = \frac{1}{8\pi^2\hbar^3}\left[p_F E_F(2p_F^2c^2 + m^2c^4) - (mc^2)^4 \ln\left(\frac{p_F c + E_F}{mc^2}\right)\right]} \tag{38}$$

4 Average Energy per Particle

4.1 Definition

The average energy per particle $\langle E\rangle$ is the ratio of total energy to total number of particles:

$$\langle E\rangle = \frac{\mathcal{E}}{N} = \frac{\varepsilon}{n} \tag{39}$$

4.2 Explicit Expression

Substituting our expressions for ε (from Eq. 45) and n (from Eq. 9):

$$\langle E\rangle = \frac{1}{8\pi^2\hbar^3}\left[p_F E_F(2p_F^2c^2 + m^2c^4) - (mc^2)^4 \ln\left(\frac{p_F c + E_F}{mc^2}\right)\right] \times \frac{3\pi^2\hbar^3}{p_F^3} \tag{40}$$

The $\pi^2\hbar^3$ terms cancel:

$$\langle E\rangle = \frac{3}{8p_F^3}\left[p_F E_F(2p_F^2c^2 + m^2c^4) - (mc^2)^4 \ln\left(\frac{p_F c + E_F}{mc^2}\right)\right] \tag{41}$$

This is the final boxed result:

$$\boxed{\langle E \rangle = \frac{3}{8p_F^3}\left[p_F E_F(2p_F^2c^2 + m^2c^4) - (mc^2)^4 \ln\left(\frac{p_F c + E_F}{mc^2}\right)\right]} \tag{42}$$

where $E_F = \sqrt{p_F^2c^2 + m^2c^4}$ is the Fermi energy.

4.3 Limiting Cases

4.3.1 Ultra-Relativistic Limit ($p_F \gg mc$)

When $p_F c \gg mc^2$, we have $E_F \approx p_F c$. The first term in the bracket becomes:

$$p_F E_F(2p_F^2c^2 + m^2c^4) \approx p_F \cdot p_F c \cdot 2p_F^2c^2 = 2p_F^4c^3 \tag{43}$$

The logarithmic term becomes negligible compared to this. Therefore:

$$\langle E \rangle \approx \frac{3}{8p_F^3} \cdot 2p_F^4c^3 = \frac{3}{4}p_F c = \frac{3}{4}E_F \tag{44}$$

4.3.2 Non-Relativistic Limit ($p_F \ll mc$)

When $p_F \ll mc$, we expand $E_F \approx mc^2 + \frac{p_F^2}{2m}$. After careful expansion and retaining leading terms:

$$\langle E \rangle \approx mc^2 + \frac{3}{5}\frac{p_F^2}{2m} \tag{45}$$

This recovers the familiar non-relativistic result: the average kinetic energy is $\frac{3}{5}$ of the Fermi kinetic energy.

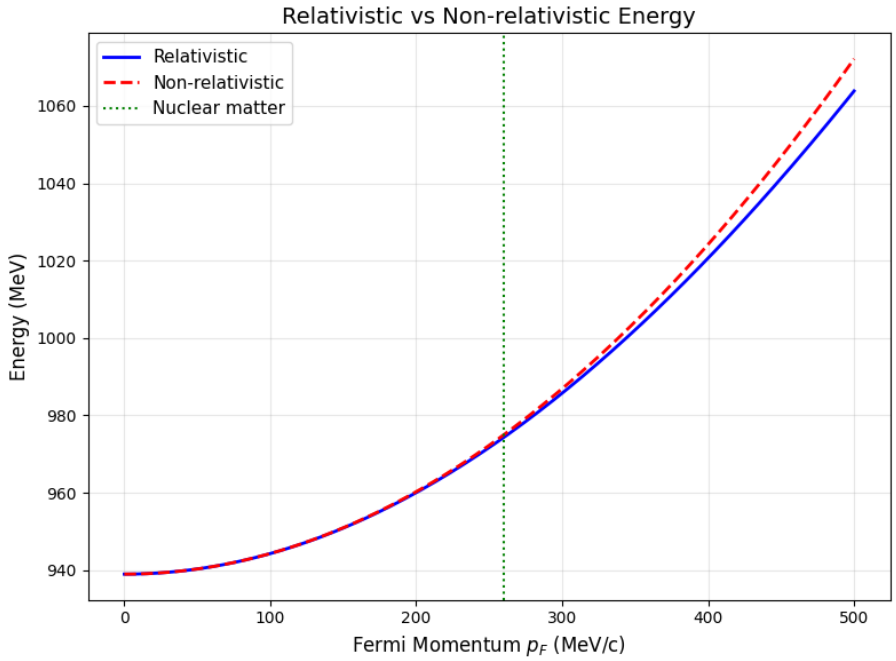


Figure 4: Comparison of average energy in relativistic (solid) and non-relativistic (dashed) limits as a function of Fermi momentum. The curves diverge significantly for $p_F > 0.3mc$.

4.4 Nuclear Matter Application

For nuclear matter with $p_F = 260$ MeV/c and $m_N c^2 = 939$ MeV:

$$E_F = \sqrt{(260)^2 + (939)^2} \text{ MeV} \approx 975 \text{ MeV} \tag{46}$$

$$\langle E \rangle \approx 939 + 36 = 975 \text{ MeV} \tag{47}$$

The average kinetic energy per nucleon is:

$$\langle E_{\text{kin}} \rangle = \langle E \rangle - mc^2 \approx 36 \text{ MeV} \tag{48}$$

This agrees well with empirical nuclear data and explains the observed binding energies in nuclei.

5 Summary and Physical Interpretation

5.1 Key Results

We have derived the following expressions for a relativistic degenerate Fermi gas:

1. **Number Density:**
$$n = \frac{p_F^3}{3\pi^2\hbar^3} \tag{49}$$

2. **Energy Density:**
$$\varepsilon = \frac{1}{8\pi^2\hbar^3}\left[p_F E_F(2p_F^2c^2 + m^2c^4) - (mc^2)^4 \ln\left(\frac{p_F c + E_F}{mc^2}\right)\right] \tag{50}$$

3. **Average Energy:**
$$\langle E \rangle = \frac{3}{8p_F^3}\left[p_F E_F(2p_F^2c^2 + m^2c^4) - (mc^2)^4 \ln\left(\frac{p_F c + E_F}{mc^2}\right)\right] \tag{51}$$

where $E_F = \sqrt{p_F^2c^2 + m^2c^4}$ is the Fermi energy.

5.2 Physical Insights

- **Relativistic Corrections:** The relativistic treatment correctly accounts for the fact that nucleons in the nucleus have momenta comparable to their rest mass. For $p_F = 260$ MeV/c and $mc = 940$ MeV/c, the ratio $p_F/(mc) \approx 0.28$ indicates that relativistic effects are non-negligible.

- **Energy Components:** The energy density contains both kinetic energy contributions and the rest mass energy of the nucleons. The logarithmic term represents the relativistic correction to the integral.

- **Nuclear Predictions:** For nuclear matter, these expressions predict:
  - Fermi momentum: $p_F \approx 260$ MeV/c

  - Fermi energy: $E_F \approx 975$ MeV
  - Average kinetic energy: $\langle E_{\text{kin}} \rangle \approx 36$ MeV

- **Nuclear Properties:** The model successfully explains nuclear saturation density, binding energy per nucleon (~ 8 MeV), incompressibility of nuclear matter, and nuclear well depth (~ 40 MeV).

5.3 Comparison with Non-Relativistic Model

In the non-relativistic limit, $E = mc^2 + \frac{p^2}{2m}$, leading to:

$$\varepsilon_{\text{NR}} = nmc^2 + \frac{3}{5} n \frac{p_F^2}{2m} \tag{52}$$

$$\langle E \rangle_{\text{NR}} = mc^2 + \frac{3}{5} \frac{p_F^2}{2m} \tag{53}$$

The relativistic corrections become important when $p_F \gtrsim 0.1mc$, which is satisfied in nuclear matter.

5.4 Applications and Limitations

This model forms the foundation for understanding nuclear structure, equation of state calculations, astrophysical modeling of neutron stars, and advanced nuclear models including Hartree-Fock theories and relativistic mean-field theory.

Limitations include neglecting nucleon-nucleon interactions, not accounting for shell structure, ignoring pairing correlations, and treating protons and neutrons independently.

— End of Assignment —

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PHL509 – Nuclear and Particle Physics

Assignment-1

Group: Mechatronics
Captain: Shivam Singh
Date: 16 February 2026

1 Nuclear Model: Degenerate Fermi Gas Model

The nucleus can be modeled as a system of interacting nucleons. To simplify the many-body problem, the nucleus is approximated as a **degenerate Fermi gas** of non-interacting fermions confined within a finite volume.

A **Fermi gas** is a collection of many identical fermions (half-integer spin particles such as electrons and neutrons) that obey the **Pauli exclusion principle** and **Fermi-Dirac statistics**. At high density or low temperature, the available low-energy quantum states become filled up to a maximum energy called the **Fermi energy** (E_F).

Since nucleons are fermions , they obey Fermi–Dirac statistics and the Pauli exclusion principle. At zero temperature ($T = 0$), all momentum states are filled up to a maximum momentum called the **Fermi momentum** (p_F).

To understand this model, the description is extended from a one-body framework to a many-body framework, in which the nucleus is treated as a collection of independent particles occupying quantum states up to the Fermi surface.

2 One Body Description

For a single particle of mass m, the physical quantities are defined as follows:

$$\begin{aligned} \text{Mass:} \quad & m \\ \text{Position:} \quad & x(t) \\ \text{Velocity:} \quad & v(t) \\ \text{Momentum:} \quad & p = mv \\ \text{Kinetic Energy:} \quad & E_K = \frac{1}{2}mv^2 = \frac{p^2}{2m} \end{aligned}$$

The state of a single particle can be represented in **phase space** using the coordinates (x, p). For example, in simple harmonic motion (SHM), the phase-space trajectory is an ellipse.

3 Many Body Description

In a gas containing many particles, the system is described using macroscopic quantities such as pressure P, volume V, and temperature T.
From classical thermodynamics, the ideal gas law is given by

$$\frac{PV}{T} = \text{constant} = R,$$

where R is the universal gas constant defined as

$$R = N_A k_B.$$

Here, $N_A \sim 10^{23}$ is the Avogadro number, and $k_B = 1.38 \times 10^{-23}\,\mathrm{J\ K^{-1}}$ is the Boltzmann constant. This relation connects microscopic physics (through k_B) to macroscopic thermodynamic quantities.

4 Phase Space for Many-Body System

For a system of N particles, each particle has three position coordinates and three momentum coordinates. Therefore, the total phase space dimension is $6N$. Thus, the system can be represented in a $6N$-dimensional phase space consisting of $3N$ position axes and $3N$ momentum axes. Macroscopic quantities arise from suitable summation over microscopic states:

$$\text{Microscopic quantities} \longrightarrow \text{Summation} \longrightarrow \text{Macroscopic quantities.}$$

$$\sum 1 = N,$$
$$\sum E_K = U,$$
$$\sum \frac{pv}{3} = PV.$$

5 Statistical Mechanics Description

In statistical mechanics, summation over states is replaced by integration in phase space:

$$\sum \quad \longrightarrow \quad \int \frac{d^3x\, d^3p}{h^3}$$

Thus,

$$\Sigma \equiv \int \frac{d^3x\, d^3p}{h^3}\, f(p)$$

where $f(p)$ is the distribution function.
For a completely degenerate Fermi gas at $T = 0$:

$$f(p) = \begin{cases} 1, & p \le p_F \\ 0, & p > p_F \end{cases}$$

6 Relativistic Degenerate Fermi Gas

The relativistic energy dispersion relation is:

$$E(p) = \sqrt{p^2c^2 + m^2c^4} \tag{1}$$

For Non-Relativistic case:

$$E(p) = \frac{p^2}{2m} \tag{2}$$

In the ultra-relativistic limit:

$$m \to 0$$

$$E(p) = pc \tag{3}$$

7 Quantum Statistics, Number Density, Energy Density and Average Energy

For a system in thermal and particle equilibrium, the thermodynamic potential is:

$$\Phi = -PV \tag{4}$$

In statistical mechanics:

$$\Phi = -kT \ln Z_G \tag{5}$$

where:

- Z_G = grand partition function
- k = Boltzmann constant
- T = temperature
- $\beta = \frac{1}{kT}$

For independent single-particle states:

$$\Phi = -kT \sum_i \ln Z_i \tag{6}$$

For Fermions:

$$Z_i = 1 + e^{-\beta(\epsilon_i - \mu)} \tag{7}$$

where ϵ is energy of the i-th single-particle state. and μ is chemical potential.

$$\Phi = -kT \sum_i \ln \left[1 + e^{-\beta(\epsilon_i - \mu)}\right] \tag{8}$$

In general, it can be written:

$$\Phi = -\frac{kT}{\eta} \sum_i \ln \left[1 + \eta e^{-\beta(\epsilon_i - \mu)}\right] \tag{9}$$

where

$$\eta = \begin{cases} +1 & \text{Fermions} \\ -1 & \text{Bosons} \end{cases}$$

Replacing summation by integration in phase space:

$$\Phi = -\frac{kT}{\eta} \int \frac{V d^3p}{h^3} \ln \left[1 + \eta e^{-\beta(\epsilon - \mu)}\right] \tag{10}$$

The particle number is obtained from:

$$N = -\left(\frac{\partial \Phi}{\partial \mu}\right)_{T,V} \tag{11}$$

Differentiating:

$$N = \frac{1}{\eta} \sum_i \frac{\eta e^{-\beta(\epsilon_i - \mu)}}{1 + \eta e^{-\beta(\epsilon_i - \mu)}} \tag{12}$$

Simplifying:

$$N = \sum_i \frac{1}{e^{\beta(\epsilon_i - \mu)} + \eta} \tag{13}$$

In continuum form:

$$N = \int \frac{V d^3p}{h^3} \frac{1}{e^{\beta(\epsilon-\mu)} + \eta} \tag{14}$$

$$N = \int \frac{V d^3p}{h^3} f_0 \tag{15}$$

Thus the occupation probability is:

$$f_0(p) = \frac{1}{e^{\beta(\epsilon-\mu)} + \eta} \tag{16}$$

For Fermi–Dirac distribution function:

$$f_0(p) = \frac{1}{e^{\beta(\epsilon-\mu)} + 1} \tag{17}$$

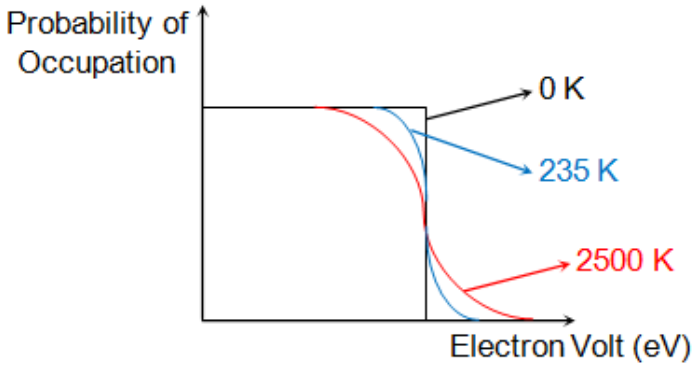


Figure 1: Fermi-Dirac Distribution Function at Different Temperatures

The total internal energy is obtained as:

$$U = \sum \epsilon f_0 \tag{18}$$

In continuum form:

$$U = \frac{V}{h^3} \int d^3p\, \epsilon\, f_0(\epsilon) \tag{19}$$

For fermions:

$$U = \frac{V}{h^3} \int d^3p \frac{\epsilon}{e^{\beta(\epsilon-\mu)} + 1} \tag{20}$$

At $T = 0$, all states are filled up to the Fermi momentum p_F.
Including spin degeneracy g, using equation 15.

$$\frac{N}{V} = n = g \int_0^{p_F} \frac{d^3p}{h^3} \tag{21}$$

Using

$$d^3p = 4\pi p^2 dp$$

$$n = g \frac{4\pi}{h^3} \int_0^{p_F} p^2 dp \tag{22}$$

$$\boxed{n = \frac{4\pi g}{3h^3} p_F^3} \tag{23}$$

Hence,

$$\boxed{p_F = \left(\frac{3h^3 n}{4\pi g}\right)^{1/3}} \tag{24}$$

Ultra-relativistic limit ($m \to 0$):
Using equation 3.

$$\boxed{n = \frac{4\pi g E_F^3}{3h^3 c^3}} \tag{25}$$

Energy density:

$$\mathcal{E} = g \int_0^{p_F} \frac{d^3 p}{h^3} E(p) \tag{26}$$

Ultra-relativistic limit ($m \to 0$):
Using equation 3.

$$\mathcal{E} = g \frac{4\pi}{h^3} \int_0^{p_F} pcp^2 dp \tag{27}$$

$$= g \frac{4\pi c}{h^3} \int_0^{p_F} p^3 dp \tag{28}$$

$$\mathcal{E} = g \frac{\pi c}{h^3} p_F^4 \tag{29}$$

$$\boxed{\mathcal{E} = g \frac{\pi c}{h^3 c^3} E_F^4} \tag{30}$$

Average Energy per Particle

$$\langle E \rangle = \frac{\mathcal{E}}{n} \tag{31}$$

Using equation 25 and 30:

$$\boxed{\langle E \rangle = \frac{3}{4} E_F} \tag{32}$$

Pressure is given by:

$$P = g \int_0^{p_F} \frac{d^3 p}{h^3} \frac{1}{3} pv \tag{33}$$

Ultra-relativistic limit ($m \to 0$):

$$v = c$$

$$P = g \frac{4\pi c}{3h^3} \int_0^{p_F} p^3 dp \tag{34}$$

$$\boxed{P = g \frac{\pi c}{3h^3} p_F^4} \tag{35}$$

Substituting p_F in terms of n:

$$\boxed{P = \frac{\pi g c}{3h^3} \left(\frac{3h^3 n}{4\pi g}\right)^{4/3}} \tag{36}$$

$$\boxed{K = \frac{\pi g c}{3h^3} \left(\frac{3h^3}{4\pi g}\right)^{4/3}} \tag{37}$$

Hydrodynamical Equilibrium and Stability of White Dwarfs

ME-2

February 15, 2026

1 Question

Derive and describe the hydrodynamical equilibrium used to explain the stability of white dwarfs.

2 Solution

2.1 Introduction

White dwarfs are compact stellar remnants supported against gravitational collapse not by thermal pressure, but by **electron degeneracy pressure**. This pressure arises from the Pauli Exclusion Principle and Fermi–Dirac statistics. Since degeneracy pressure depends primarily on density rather than temperature, white dwarfs can remain stable even after they cool.

2.2 Hydrostatic Equilibrium

Consider a spherical shell of radius r, thickness dr, and density $\rho(r)$.

Mass of the shell

$$dm = 4\pi r^2 \rho(r)\, dr \tag{1}$$

Gravitational force

The inward gravitational force on the shell is

$$F_g = -\frac{GM(r)}{r^2}\, dm \tag{2}$$

Substituting for dm:

$$F_g = -\frac{GM(r)}{r^2}\left(4\pi r^2 \rho(r)\, dr\right) \tag{3}$$

Pressure force

The outward force due to the pressure gradient is

$$F_p = -\frac{dP}{dr}\, 4\pi r^2\, dr \tag{4}$$

Condition for Equilibrium

For hydrostatic equilibrium:

$$F_g + F_p = 0 \tag{5}$$

This yields the fundamental equation:

$$\boxed{\frac{dP}{dr} = -\frac{GM(r)\rho(r)}{r^2}} \tag{6}$$

Together with the mass continuity equation:

$$\frac{dM}{dr} = 4\pi r^2 \rho(r) \tag{7}$$

These two equations determine the stellar structure.

2.3 Equation of State: Degenerate Electron Gas

White dwarf stability depends on how pressure varies with density. For a completely degenerate electron gas, the equation of state is polytropic:

$$P = K\rho^{\gamma} = K\rho^{1+1/n} \tag{8}$$

where n is the polytropic index.

2.3.1 Non-Relativistic Degeneracy ($n = 3/2$)

For $v \ll c$:

$$P_{NR} = \frac{h^2}{20m_e}\left(\frac{3}{\pi}\right)^{2/3}\left(\frac{\rho}{\mu_e m_H}\right)^{5/3} \tag{9}$$

Here $\gamma = 5/3$ and $n = 3/2$.
In this regime:

$$R \propto M^{-1/3} \tag{10}$$

As mass increases, radius decreases, but stable equilibrium solutions exist.

2.3.2 Relativistic Degeneracy ($n = 3$)

When electrons become ultra-relativistic:

$$P_{ER} = \frac{hc}{8}\left(\frac{3}{\pi}\right)^{1/3}\left(\frac{\rho}{\mu_e m_H}\right)^{4/3} \tag{11}$$

Here $\gamma = 4/3$ and $n = 3$.

In this case, equilibrium mass becomes independent of radius, leading to the **Chandrasekhar mass limit**:

$$M_{Ch} = \frac{5.83}{\mu_e^2} M_{\odot} \approx 1.44 M_{\odot} \tag{12}$$

2.4 Lane–Emden Equation

Combining hydrostatic equilibrium and mass conservation gives

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dP}{dr}\right) = -4\pi G\rho \tag{13}$$

Substituting the polytropic relation and defining

$$\rho = \rho_c\theta^n, \quad r = a\xi \tag{14}$$

we obtain the Lane–Emden equation:

$$\frac{1}{\xi^2}\frac{d}{d\xi}\left(\xi^2\frac{d\theta}{d\xi}\right) = -\theta^n \tag{15}$$

Solutions of this equation determine the internal structure of white dwarfs.

2.5 Energy Argument and Stability

The total energy of a white dwarf is

$$E = U + \Omega \tag{16}$$

where:

- U = internal (degeneracy) energy
- Ω = gravitational potential energy

For non-relativistic degeneracy:

$$U \propto \frac{1}{R^2}, \quad \Omega \propto -\frac{1}{R}$$

A stable minimum energy configuration exists.

For relativistic degeneracy:

$$U \propto \frac{1}{R}, \quad \Omega \propto -\frac{1}{R}$$

The total energy approaches zero, leading to marginal stability. Thus $n = 3$ corresponds to the maximum stable mass.

2.6 Conclusion

White dwarfs remain stable due to hydrostatic equilibrium between gravity and electron degeneracy pressure.

For $n = 3/2$ (non-relativistic case), stable configurations exist with $R \propto M^{-1/3}$.

For $n = 3$ (relativistic case), the equilibrium mass approaches the Chandrasekhar limit ($\approx 1.44M_\odot$). Beyond this limit, hydrostatic equilibrium fails and gravitational collapse occurs.